Soft Condensed Matter: An Introduction (2025)

Lecture 5: Elastic vs Viscous vs Viscoelastic Behavior

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Key Topics

How Materials Deform

Contrasting three fundamental behaviors:

1. **Elastic Solids** - Store energy, recover shape
2. **Viscous Fluids** - Dissipate energy, flow continuously
3. **Viscoelastic Materials** - Combine both behaviors

1. Elastic (Hookean) Behavior

Defining Characteristics

Hooke's Law:

$$\sigma = E \cdot \varepsilon \tag{1}$$

Where:

- σ = stress (force per unit area)
- E = Young's modulus (material stiffness)
- ε = strain (fractional deformation)

Key Properties:

- Stress proportional to **strain** (not rate)
- Deform under load but **store energy**
- **Recover original shape** when load removed
- Energy stored elastically (like a spring)

Examples

- Spring
- Rubber band (small extension)
- Steel beam (small deformations)
- Hard plastic below glass transition temperature

Stress-Strain Curves

Linear Elastic Regime:

- Straight line
- Slope = Young's modulus E
- Higher E → more stress needed for same strain (stiffer)

Shear Modulus:

- For shear deformations: $\tau = G \cdot \gamma$
- G = shear modulus
- Similar concept, different geometry

Typical Moduli Values

- **Steel:** E ~ 200 GPa (very stiff)
- **Rubber:** E ~ 1 MPa (soft)
- **Silicone gel:** E ~ 50 kPa (very soft)
- **Brain tissue:** E ~ 1 kPa (extremely soft)

2. Viscous (Newtonian) Behavior

Defining Characteristics

Newton's Law of Viscosity:

$$\tau = \eta \cdot \dot{\gamma} \tag{2}$$

Where:

- τ = shear stress
- η = viscosity (Pa·s)
- $\dot{\gamma}$ = shear rate (strain rate)

Key Properties:

- Stress proportional to **strain rate** (not strain itself)
- Flows continuously under any applied stress
- **Dissipates energy** as heat (irreversible)
- No shape recovery when stress removed

Examples

- Water ($\eta \approx 1$ mPa·s)
- Honey ($\eta \approx 10$ Pa·s, ~10,000× water)
- Glycerol ($\eta \approx 1$ Pa·s)
- Motor oil

Dimensional Analogy

Property	Elastic	Viscous
Relation	$\sigma = E \cdot \varepsilon$	$\tau = \eta \cdot \dot{\gamma}$
Units	$Pa = N/m^2$	Pa·s
Response to	Strain	Strain rate
Energy	Stored	Dissipated

Behavior Under Load

- If you stop applying stress to viscous fluid:
  - Deformation continues until stress is zero
  - No recovery
  - Fluid remains in deformed state

3. Viscoelastic Behavior

Combined Characteristics

Definition: Materials exhibiting **both** elastic and viscous behaviors

- Deform like elastic solid on **short timescales**
- Flow like viscous fluid over **long times**
- Behavior depends on observation timescale

Common Viscoelastic Materials

- Polymers (quintessential examples)
- Biological tissues (cartilage, muscle, skin)
- Gels
- Silly putty (classic demo material)
- Many food products (dough, cheese, etc.)

Spring-Dashpot Models

Maxwell Model

Configuration: Spring and dashpot in **series**

Represents: Fluid with memory

Behavior:

- Can relax stress over time
- If sudden strain applied and held:
  - Initially: High stress (spring compressed)

  - Over time: Stress gradually decays (dashpot flows)
  - Eventually: Stress → 0 (full relaxation)

Equation:

$$\dot{\varepsilon} = \frac{\dot{\sigma}}{E} + \frac{\sigma}{\eta} \tag{3}$$

Relaxation Time:

$$\tau_M = \frac{\eta}{E} \tag{4}$$

Kelvin-Voigt Model

Configuration: Spring and dashpot in **parallel**

Represents: Solid that creeps under load

Behavior:

- Under sudden load:
  - No instantaneous elastic jump (dashpot resists)
  - Gradual spring deformation (creep)
  - Approaches equilibrium asymptotically
- When load removed:
  - Slow recovery back to original shape

Equation:

$$\sigma = E\varepsilon + \eta\dot{\varepsilon} \tag{5}$$

Retardation Time:

$$\tau_K = \frac{\eta}{E} \tag{6}$$

Stress Relaxation and Creep

Stress Relaxation

Experiment:

1. Apply sudden fixed strain
2. Hold constant
3. Measure stress vs time

Results:

- **Elastic material:** Constant stress (no decay)
- **Viscoelastic (Maxwell):** Stress decays exponentially

$$\sigma(t) = \sigma_0 e^{-t/\tau_M} \tag{7}$$

- **Time constant** τ_M = relaxation time

Creep

Experiment:

1. Apply constant stress
2. Measure strain vs time

Results:

- **Elastic:** Immediate strain, then constant
- **Viscous:** Strain increases linearly forever
- **Viscoelastic (Kelvin-Voigt):**
  - Immediate dashpot resistance
  - Gradual approach to equilibrium
  -
$$\varepsilon(t) = \frac{\sigma_0}{E}(1 - e^{-t/\tau_K}) \tag{8}$$

Relaxation Time Concept

Definition

Relaxation Time (τ): Characteristic time for material to transition from elastic to viscous response

Physical Meaning:

- Maxwell time: $\tau = \eta/E$
- Time for stress to decay to $1/e$ of initial value
- Separates elastic (short-term) and viscous (long-term) regimes

Time-Scale Dependence

Fast Observation ($t \ll \tau$):

- Material behaves **elastic** (solid-like)
- Dashpot "frozen" (hasn't had time to flow)
- Spring dominates

Slow Observation ($t \gg \tau$):

- Material behaves **viscous** (liquid-like)
- Spring equilibrated
- Dashpot flows freely

Application to Silly Putty:

- $\tau \sim$ few seconds
- Quick poke ($t \ll \tau$): Bounces like ball (elastic)
- Slow pull ($t \gg \tau$): Flows like liquid (viscous)

Examples and Case Studies

Silly Putty Demonstration

Classic Demo:

Fast Impact (High Strain Rate):

1. Roll into ball
2. Drop from height
3. **Bounces** like elastic ball
4. Behavior: Solid-like ($t \ll \tau$)

Slow Deformation (Low Strain Rate):

1. Take same putty
2. Slowly pull apart
3. **Flows** and strands out like liquid
4. Can form thin filament that necks
5. Behavior: Liquid-like ($t \gg \tau$)

Very Fast (Brittle Failure):

- Quick yank → snaps (brittle behavior)
- Elastic domination with insufficient time for viscous flow

Metal vs Polymer Comparison

Metal Spring:

- Apply weight briefly
- Springs back **immediately**
- Pure elastic response
- Negligible viscous component

Viscoelastic Polymer Strip:

- Apply same weight
- Might slowly rebound
- Or not fully recover if short time
- Clear viscoelastic behavior

Pitch Drop Experiment

Famous Long-Term Experiment:

Short Timescale (Impact):

- Block of pitch (tar)

- Hit with hammer → **shatters**
- Feels solid, elastic

Long Timescale (Years):

- Over ~10 years, drops form and fall out
- **Flows** like extremely viscous liquid
- Demonstrates extreme relaxation time (years!)

Lesson: Time scale of observation matters!

Quantitative Example

Dynamic Testing:

- **Water:** Phase angle $\delta \approx 90°$ (all viscous, no elastic storage)
- **Rubber:** Phase angle $\delta \approx 0°$ (all elastic storage)
- **Silly putty:** Phase angle $\delta \approx 45°$ at intermediate frequencies (half-half)

Units Reminder:

- Elastic modulus: Pa
- Viscosity: Pa·s
- Example values:
  - Rubber E ~ 1 MPa
  - Honey η ~ 10 Pa·s
  - Silly putty τ ~ seconds

Learning Outcomes

Students will be able to:

1. **Differentiate elastic vs viscous response**
   - State defining constitutive relations (Hooke's law vs Newton's viscosity law)
   - Qualitatively describe behavior under load
2. **Give examples of each type**
   - Elastic: Steel, hard plastic below T_g, rubber (small strains)
   - Viscous: Water, oil, glycerin
   - Viscoelastic: Silly putty, cartilage, polymer melt
   - Justify classifications
3. **Understand viscoelastic models**
   - Maxwell model: Initial stress from spring, then decays as dashpot flows
   - Kelvin-Voigt: Sudden stress → dashpot resistance + gradual spring deformation
   - Conceptual understanding (no ODE solving required)
4. **Interpret storage vs loss qualitatively**

- Viscoelastic material has:
  - **Storage modulus** (elastic part, energy stored)
  - **Loss modulus** (viscous part, energy dissipated)
- More elastic → bounces back (stores energy)
- More viscous → damps motion (loses energy as heat)

5. **Apply time-scale reasoning**
   - Predict material behavior based on deformation rate
   - Quick stretch → stiff (elastic)
   - Slow stretch → soft/flowy (viscous)
   - Articulate relaxation time concept

Discussion Questions

1. Identify Behaviors

Scenarios to classify:

- **Bowling ball impacts cement floor and rebounds**
  - Mostly elastic (some energy lost)
- **Person slowly sinking into memory foam mattress**
  - Viscoelastic (slow deformation over time)
- **Toothpaste extruded from tube**
  - Mostly viscous flow (with yield stress)
- **Guitar string vibrating**
  - Elastic

Task: Students debate and justify answers

2. Spring vs Dashpot Analogy

Questions:

- If you stretch Maxwell material (spring+dashpot series) and hold, what happens to stress?
  - Initially: High stress from spring
  - Over time: Gradually decays (dashpot relieves it)
  - **Stress relaxation**
- If suddenly released after long hold, will it return to original length?
  - No – dashpot has flowed (permanent deformation)
  - Only spring part returns
  - Some deformation remains
- How does this differ from Kelvin-Voigt under sudden load?

3. Energy Storage vs Loss

Scenario: Two materials feel similarly stiff when tapped, but one gets warm when bent repeatedly, the other doesn't.

Question: What does this indicate?

- Warming material: Dissipating energy → higher **loss modulus**
- Non-warming: Stores-returns energy → higher **storage modulus**

Examples:

- Rubber bands heating up after stretching
- Car tires heating up (viscoelastic damping)

4. Design Considerations

Shoe Sole for Runners:

- More elastic or more viscous?
- **Discussion:**
  - Elastic: Returns energy (spring in step)
  - Viscous: Absorbs energy (shock absorption)
  - Likely want mix (viscoelastic)
  - Different answers possible
  - Shows engineering balance

Seismic Dampers:

- Should be highly **viscous**
- Dissipate earthquake energy
- Not elastic (would spring back and forth)
- Connects material behavior to function

Key Concepts to Remember

Elastic Behavior

- **Relation:** $\sigma = E \cdot \varepsilon$
- **Energy:** Stored (recoverable)
- **Response:** Immediate
- **Examples:** Springs, metals, rubber (small strain)

Viscous Behavior

- **Relation:** $\tau = \eta \cdot \dot{\gamma}$
- **Energy:** Dissipated as heat
- **Response:** Continuous flow
- **Examples:** Water, honey, oils

Viscoelastic Behavior

- **Models:** Maxwell (fluid-like) vs Kelvin-Voigt (solid-like)
- **Key parameter:** Relaxation time $\tau = \eta/E$
- **Time dependence:** Solid when fast, liquid when slow
- **Examples:** Polymers, silly putty, tissues

Storage vs Loss

- **Storage modulus (G′):** Elastic energy stored
- **Loss modulus (G″):** Viscous energy dissipated
- **Phase angle (δ):** $\tan \delta = G''/G'$
- Sets up next lecture on rheology!

Preparation for Next Lecture

Next lecture: **Rheology – Measuring "Softness"**

Consider:

- How do we experimentally measure these properties?
- What is a rheometer?
- How do G′ and G″ relate to oscillatory tests?
- What does "soft" really mean quantitatively?

References

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Previous Lecture: Lecture 4: Finite-Time Singularities

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